

Negative Exponent LawIndependent Set 3.04 **ANSWER KEY***Six questions, on your own, in silence. Show the step that does the work.*

1. Rewrite with a positive exponent. Do not evaluate.
- 9^{-2}

$$\frac{1}{9^2}$$

2. Find the value.
- 3^{-4}

$$\frac{1}{81}$$

Reciprocal of 3^4 , not -81 .

3. Find the value.
- $(-4)^{-2}$

$$\frac{1}{16}$$

An even power of a negative base is positive.

4. Find the value.
- $\frac{1}{10^{-3}}$

$$1,000$$

The power moves up.

5. Find the value.
- $5^2 \cdot 5^{-5}$

$$5^{-3} = \frac{1}{125}$$

$$2 + (-5) = -3.$$

6. Find the value of
- n
- .
- $2^n = \frac{1}{16}$

$$n = -4$$